Visual Difficulties Reported by Low-Vision and Nonimpaired Older Adult Drivers

Loretta Neal McGregor, Ouachita Baptist University, Arkadelphia, Arkansas, and **Alex Chaparro,** Wichita State University, Wichita, Kansas

Nonimpaired and low-vision older adults responded to a questionnaire regarding the types of visual difficulties experienced while performing daily tasks and while driving. Using the factors produced from a factor analysis as predictors, a discriminant analysis was performed to determine whether significant differences in visual problems existed between the groups. The majority of participants reported that they currently required more time than in the past to perform tasks that depended on their vision, regardless of their visual status. All participants reported experiencing significant difficulties with static and dynamic acuity, peripheral vision, illumination problems, and contrast sensitivity. Both nonimpaired and low-vision older adult drivers reported experiencing difficulty with glare, peripheral vision, and night driving. Low-vision drivers reported experiencing unique difficulties with near acuity, distant acuity, and physical obstructions. Potential applications of this research include suggestions for redesigning automobiles and highway signs for safer use and viewing.

INTRODUCTION

The majority of older adults continue to drive safely well into their 80s and 90s (Ball & Rebok, 1994; Federal Highway Administration, 1998; Johnson & Keltner, 1983; Marottoli et al., 1993). Many older drivers drive because they like the freedom of coming and going as they please, but many others continue to drive out of necessity (Orr, 1991). Data indicate that some older drivers often continue to drive because of a lack of adequate public transportation within their community and the fear of isolation or of being a burden to others if they gave up driving (Orr, 1991).

However, older drivers are often depicted in the movies and on television as slow drivers, accident prone, and a driving risk to self and others. It is true that older drivers, as a group, have more traffic accidents and more fatalities per miles driven than all other groups except young drivers; however, one should consider that a wide range of individual differences in driving capabilities exists among older drivers. Ball et al. (1998) suggested that because such wide variability does exist, only a subset of older drivers – those with severe functional impairments – increase the average risk of an accident for the entire age group.

This depiction of older drivers as a driving risk to self and others has led governing bodies and many within the public sector to question whether an upper age limit or other restrictions should be imposed on driving. Some state legislatures have considered denying drivers' licenses to individuals who exceed a particular age. Additionally, some states are considering placing restrictions (e.g., specifying under what conditions a person will be allowed to drive) on older drivers (Ball, Owsley, Sloane, Roenker, & Bruni, 1993). A few states now require an annual driver license renewal for individuals over the age of 60 (Federal Highway Administration, 1998). However, Ball et al. (1993) wrote, "policies that restrict driving privileges based solely on age or on common stereotypes of age-related declines... are scientifically unfounded" (p. 3110).

The majority of states require that drivers

Address correspondence to Loretta Neal McGregor, Department of Psychology and Counseling, P.O. Box 1560, Arkansas State University, State University, AR 72467; lmcgregor@astate.edu. **HUMAN FACTORS,** Vol. 47, No. 3, Fall 2005, pp. 469–478. Copyright © 2005, Human Factors and Ergonomics Society. All rights reserved.

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have a best-corrected visual acuity of 20/40 to obtain a nonrestricted driver's license. However, these visual standards vary from state to state: Kansas will grant a restricted driver's license to individuals with a best corrected acuity of 20/90, but California will allow an individual with a best corrected acuity of 20/200 (which is the legal definition for blindness) to obtain a restricted driver's licenses (Charman, 1985; Fishbaugh, 1995; *Kansas Driving Handbook,* 1999; Keltner & Johnson, 1987).

Only a limited number of studies have focused exclusively on the relationship between vision and driving (Ball et al., 1993; Fox, 1988; Keltner & Johnson, 1987; Kline et al., 1992; Kosnik, Sekuler, & Kline, 1990; Kosnik, Winslow, Kline, Rasinski, & Sekuler, 1988; Owsley et al., 1998). Results from many of these studies suggest, both directly and indirectly, that the vision problems that plague many older adults have some effect on their driving abilities. Kosnik et al. (1988, 1990) found that older drivers reported more difficulty with performing a variety of everyday activities (e.g., difficulty reading small print, difficulty distinguishing dark colors, eyestrain doing close work) because of their declining vision. These problems were more pronounced in older persons who had recently given up driving. Also, similar to Kosnik et al.'s (1990) results, Kline et al.'s (1992) results indicated that the vision problems experienced by current older drivers increased with age. Additionally, Kline et al. (1992) identified age-related visual declines along specific driving dimensions. Older adult drivers reported difficulty in estimating vehicle speed, viewing a dimly lit dashboard, and reading street signs. These participants also complained more about the unexpected appearance of vehicles (merging or passing) and windshield problems such as glare.

Ball et al. (1993) expanded the research parameters set forth by Kosnik et al. (1988) and Kline et al. (1992). Ball et al. (1993) examined the visual information-processing ability and other visual capabilities of adult drivers to determine whether a relationship existed between the participants' number of reported automobile accidents and their visual abilities. They examined six areas of visual function in an attempt to find a correlation between vision and crash history. The six visual capabilities examined were static visual acuity (the ability to detect details clearly in nonmoving objects), contrast sensitivity (one's ability to distinguish between an object and its background when the two resemble each other closely in color), glare, stereopsis or depth perception, visual field sensitivity (the necessary intensity of a presented object for perception), and color discrimination. The authors also measured the size of each participant's useful field of view. Ball et al. (1993) hypothesized that as the field of view decreases with age, so does the useful field of view. Unlike the researchers who preceded them, Ball et al. (1993) made an attempt to include individuals with poor vision (visual acuity worse than 20/40). Results from the study indicated that several of the visual measures were correlated with the size of an individual's useful field of view. However, the only indicators that correlated highly with crash involvement and discriminated well between crash-involved and crash-free drivers was the size of the useful field of view.

Although Kosnik et al. (1988), Kline et al. (1992), Ball et al. (1993), and others have acknowledged that vision changes as one ages, each of these studies excluded individuals with a diagnosed visual impairment, and none of them included individuals diagnosed as having low vision. Kosnik et al. (1988) and Kline et al. (1992), in particular, limited participation in their studies to individuals whose acuities were equivalent to that adopted by most states for granting an unrestricted driver's license: 20/40. Although Ball et al. (1993) included individuals with worse acuities in their study, those individuals did not have a diagnosed visual impairment. Because most studies exclude older drivers with a diagnosed visual impairment, little is known about this particular group of drivers. Therefore it is unknown if drivers who have specific visual disorders experience unique problems when compared with older drivers with nonimpaired vision.

In essence, none of these studies provided an in-depth examination of the relationship between driving and individuals in the population with poor vision. For example, what percentage of older adults diagnosed with low vision continue to drive? What specific driving tasks (if any) do low-vision drivers find particularly difficult? How do these drivers alter their driving habits to compensate for their vision loss, and how do

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their adaptive strategies compare with those used by older adult drivers who do not have a diagnosed visual impairment? Furthermore, little information is available about the type of daily visual problems experienced by older individuals who have visual acuities worse than 20/40 or those who have a diagnosed eye disease and how they compare with older adults with nonimpaired vision. Does their declining vision interfere with the simple performance of daily tasks?

Many eye diseases are related to low vision, and most of these diseases are influenced by the aging process. In order to describe the effects of age-related visual changes on driving, one must include low-vision drivers in the research. The purpose of this investigation was to obtain information about the visual experiences of low-vision older adults who continue to drive and to examine how their declining vision affects their performance of both daily and driving tasks.

METHOD

Participants

Participants were 195 low-vision and nonimpaired adult drivers and nondrivers, 50 years of age or older. All participants were volunteers. All low-vision participants met one of the following requirements: (a) a diagnosed visual impairment caused by an eye disease (e.g., glaucoma, retinitis pigmentosa, age-related macular degeneration) and/or (b) corrected visual acuity worse than 20/40. The majority of the participants were recruited from low-vision rehabilitation centers in Wichita, Kansas, and St. Louis, Missouri, and from residential retirement centers in Wichita and in Belen, New Mexico.

Materials

Each participant completed a questionnaire packet. The packet consisted of a consent form, a medical release form, and a diagnostic form, all developed by the researchers. The packet also included a copy of "Your Vision: A Questionnaire by the Vision Laboratories of Northwestern University and the University of Calgary" (Kline et al., 1992).

The diagnostic form consisted of 10 questions regarding the visual health of the participant. Information such as visual acuity and diagnosed eye disease or diseases was recorded on this form. The "Your Vision" questionnaire is composed of three distinct sections. The first section requested demographic information along with self-reported visual information. The second section required the participant to respond to 19 Likert scale questions regarding daily visual tasks. Each question had four possible responses and, depending on the wording of the questions, the response patterns ranged from *none at all* to *a lot* or *never* to *frequently.* The third and final section of the "Your Vision" questionnaire opened with questions regarding whether the respondent was currently a licensed driver. If the respondent answered no, this concluded his or her participation. If the answer was yes, the questionnaire directed the participant to respond to 26 questions regarding his or her visual experiences while driving. As in the second section, each of these questions was accompanied by four Likert-style responses and, depending on the wording of the questions, the response patterns ranged from *none at all* to *a lot* or *never* to *frequently.* This section also included questions regarding the total number of miles driven during the past year, the percentage of night driving and rush hour driving engaged in by the participant, and the type of driving environment most frequented by the participant. These questions were designed to examine the driving habits of older drivers. The questionnaire concluded with a free-response section that allowed participants to present any other information that they deemed relevant.

Procedure

Participants at the low-vision clinics completed the questionnaire packet during a scheduled visit to the clinic. Participants who did not complete the questionnaire at the clinic were provided a self-addressed return envelope in which they could mail the completed questionnaire. The questionnaires were read aloud to all participants who found it difficult to read the printed materials.

The researchers or clinic personnel completed the diagnostic forms. The information provided on these forms was taken from medical records provided by the participants' optometrist or ophthalmologist. This information was used to determine whether the participant was a nonimpaired or a low-vision driver. Once the form

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was completed, each participant was thanked and provided with written information on how to obtain the results of the study.

The completed questionnaires were divided into two groups, those completed by drivers with nonimpaired vision and those completed by drivers with low vision, for the purpose of analysis.

RESULTS

Demographic Profile

A total of 195 participants completed the vision questionnaires. The mean age of the participants was 78.54 years (SD = 8.85). Women constituted 55.9% of the sample, and 43.1% of the participants were men. Two participants did not list their gender. Additionally, 55.4% of the participants experienced some type of visual impairment, whereas 44.6% of the participants had no visual impairments. The most common type of visual impairment reported was age-related macular degeneration (ARMD). This finding is interesting to note because AMRD results in the death of retinal cells and yet 57.4% of the visually impaired participants reported that they continued to drive. A total of 72.8% of all participants continued to drive regardless of their visual status. The participants who had discontinued driving (M = 80.91 years, SD = 9.83) were significantly older, $t(80) = 2.14$, $p = .04$, than those who continued to drive (M = 77.66 years, SD = 8.30).

Driving Habits of Low-Vision and Nonimpaired Older Adults

A series of t tests were performed on the questions pertaining to driving habits. Low-vision drivers differed significantly (M = 2.08%, SD = 1.46%) from nonimpaired drivers (M = 2.48%, SD = 1.10%) in the percentage of time spent driving at night, $t(140) = 2.69$, $p = .01$. Low-vision participants reported that between 0% and 5% of their total driving occurred at night; nonimpaired drivers spent between 6% and 10% of their total driving time at night. The groups did not differ significantly in the percentage of driving done in rush hour traffic, $t(122) = 1.29$, $p = .20$, nor did they differ significantly in the type of driving environments in which they drove $t(125) = 1.42$, $p = .16$. Low-vision and nonimpaired drivers also did not differ significantly in the total number of miles driven within the past year, $t(133) = 0.15$, $p = .89$.

Factor Analysis of Daily Visual Tasks

A principal factor analysis with orthogonal varimax rotations followed by oblique promax rotations was performed using the User Oriented Factor Analytical Package (USOFAP; Burdsal, 1981). The variables analyzed were the responses given to the Likert scale questions regarding daily visual tasks. Factor extraction was based on a Scree test produced by a principal component analysis. After finding no outliers or multicollinearity, the principal factor analysis yielded five factors with a simple structure of 56.7 with a .10 hyperplane. Responses from 195 participants were included in the analysis.

Table 1 shows how well the estimated factor scores correlated with the actual factors (*Rf*s). It should be noted that although the lowest reliability produced was .89, several factors were interrcorrelated as indicated by the numerous loadings of .70 and higher. Each factor was constructed using only salient variables with loadings of .34 or higher on the factor pattern matrix. Factor 1 consisted of questions that related to the effects of reduced static acuity (e.g., reading small print). Factor 2 addressed issues related to bumping into things like doorways, walls, or people "which one did not see." These variables were related to peripheral vision. The variables that loaded on Factor 3 were related to dynamic acuity. Factor 4 addressed illumination level and problems associated with rapid changes in illumination, such as "difficulty adjusting to

TABLE 1: Correlation of Factor Scores With Factors for Daily Visual Task Questions (*Rf*s)

	Factor Scores				
Factor	1	2	3	4	5
1	*.95*	.31	.72	.70	.70
2	.30	*.91*	.37	.47	.33
3	.72	.39	*.95*	.69	.72
4	.67	.48	.67	*.91*	.63
5	.65	.33	.67	.62	*.89*

Note. Scores on diagonal in italics represent the reliability of each factor.

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bright lights when coming out of a dark place." Finally, Factor 5 dealt specifically with contrast issues (e.g., "difficulty distinguishing between dark colors").

Factor Analysis of Visual Experiences in Driving

Again, using USOFAP, a principal component analysis was performed using the Likert scale questions regarding visual experiences in driving as variables. The principal component analysis involved orthogonal varimax rotations followed by oblique promax rotations. Responses from only those participants who were current drivers were included in the analysis. Factor extraction was based on the results of a Scree test. The principal component analysis did not indicate the presence of any outliers, nor was there multicollinearity. A total of six factors were extracted with a simple structure of 61.1 with a .10 hyperplane. All missing values were handled by pairwise deletion.

The correlation of factor scores with factors (*Rf*s) and the reliability of each factor appear in Table 2. The reliabilities produced for these factors ranged from .98 to .83 and showed less intercorrelation among scores than did the variables taken from the daily vision questionnaire. Variables that loaded .34 or higher on the factor pattern matrix were used to produce the six factors. Variables related to near vision at night loaded on Factor 1. Questions pertaining to peripheral vision loaded on Factor 2. Factor 3 consisted of questions that related to problems associated with glare during night driving. Variables loading onto Factor 4 were related to problems associated with visual obstructions. Questions loading on Factor 5 dealt with problems associated with motion perception. Finally, Factor 6 was related to problems associated with distance acuity.

TABLE 2: Correlation of Factor Scores With Factors for Experiences in Driving Questions (*Rf*s)

	Factor Scores					
Factor	1	2	3	4	5	6
1	*.98*	.11	.28	.25	.17	.58
2	.10	*.90*	.30	.29	.43	.45
3	.27	.32	*.96*	.06	.35	.44
4	.22	.28	.06	*.87*	.34	.26
5	.14	.40	.31	.32	*.83*	.29
6	.52	.44	.41	.26	.30	*.88*

Note. Scores on diagonal in italics represent the reliability of each factor.

Discriminant Analysis of Daily Visual Tasks

The factors produced from the daily visual task questions were used as predictors in a discriminant analysis in an attempt to examine any dimensions on which low-vision and nonimpaired older adults might differ. Of the original 195 participants who responded to the daily visual task questionnaire, data from 2 were excluded from the discriminant analysis as a result of incomplete data, resulting in a total 193 cases. Data analysis resulted in the production of one statistically significant linear discriminant function (LDF), $\chi^2(5, N = 129.00)$, $p < .001$, with a Wilks's lambda of .50.

As indicated by Table 3, all of the predictors were found to discriminate between low-vision and nonimpaired participants. Low-vision older adults generally experienced more difficulty with contrast, static and dynamic acuity, illumination, and peripheral vision than did nonimpaired older adults. The correlations between discriminating variables and the function, also located in Table 3, shows the degree to which each predictor correlated with the function. Although each predictor contributed significantly to the LDF, two of the correlations were very low (e.g., peripheral vision and contrast both at .24), suggesting a much lower contribution in comparison with the other predictors. Static acuity contributed the most to the LDF, with a correlation of .92. The pooled within-group correlation matrix also shows that some degree of intercorrelation existed among the predictors (see Table 4). Unfortunately, these matrix scores suggest that the contributions of some predictors overlapped. Further analyses suggested that static acuity was a mediating predictor for both dynamic acuity and illumination. Finally, the LDF correctly classified 83.42% of the participants. More specifically, 12.9% of the nonimpaired participants were misclassified as being low-vision, and 19.4% of the low-vision participants were misclassified as having no visual impairments.

The patterns of the standardized canonical discriminant function coefficients (see Table 3),

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TABLE 3: Summary of Discriminant Analysis for Daily Task Predictors[a]

Predictor Variable	Standardized Function Coefficients	Unstandardized Function Coefficients	Pooled Within-Group Correlation Between Variables and Functions	Wilks's Lambda
Static acuity	1.39**	1.96	.92	.55
Peripheral vision	0.21*	0.23	.24	.95
Dynamic acuity	–0.42**	–0.48	.44	.84
Illumination	–0.33**	–0.40	.43	.84
Contrast	0.02**	0.03	.24	.81
Constant		–0.01		

[a] χ^2 (5, N = 193) = 129.00, $p < .001$, Wilks's lambda = .50.
*$p < .005$. **$p < .001$.

or standardized weights, suggest that static acuity seemed to be the most significant problem experienced by the majority of low-vision older adults. Static acuity, more than any other predictor, uniquely distinguished the low-vision participants from the nonimpaired participants. Conversely, problems with dynamic acuity and illumination were uniquely representative of the visual problems experienced by the nonimpaired participants. The group centroids and the mean discriminant scores of each predictor (see Table 5) illustrate the significant differences between the low-vision and nonimpaired older adults, especially when comparing the groups on problems associated with static and dynamic acuity and illumination.

Discriminant Analysis of Experiences in Driving

The factors produced from the questions about experiences while driving were used as predictors in a discriminant analyses in an attempt to examine any dimensions on which low-vision and nonimpaired older drivers might differ. The analysis resulted in one significant linear discriminant function, $\chi^2(6, N = 21.71)$, $p < .01$, with a Wilks's lambda of .80. Interestingly, Table 6 shows that only three of the six predictors contributed significantly to the LDF. Because near acuity, distance acuity, and physical obstruction all contributed significantly to the LDF, it is not surprising that they also had the highest correlations between the predictors and the function (see Table 7). The pooled within-group correlation matrix (see Table 7) shows that all of the predictors had good internal consistency with little or no intercorrelation among them. Finally, the LDF correctly classified 67.96% of the participants. More specifically, 26.9% of the nonimpaired participants were misclassified as having low vision and 41.7% of the low-vision participants were misclassified as having no visual impairment. On closer inspection, the standardized canonical discriminant function coefficients (see Table 6), or standardized weights, indicated that near acuity, physical obstructions, and distance acuity were problems that contribute uniquely to the driving experiences of low-vision older adults.

Specifically, nonimpaired drivers reported motion perception as being most troublesome to the driving task. However, it did not contribute significantly to the LDF. The results in Table 8 indicate that low-vision drivers reported

TABLE 4: Pooled Within-Group Correlation Matrix for Daily Visual Task LDF

Predictor	1	2	3	4	5
Dynamic acuity	1.00				
Contrast	.71	1.00			
Illumination	.68	.63	1.00		
Peripheral vision	.36	.31	.48	1.00	
Static acuity	.72	.66	.68	.24	1.00

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TABLE 5: Group Means and Standard Deviations for Daily Visual Tasks LDF

	Nonimpaired		Low Vision	
Predictors	*M*	*SD n*(193)	*M*	*SD n*(193)
Contrast	–0.42	0.74	0.35	0.85
Dynamic acuity	–0.42	0.79	0.40	0.93
Illumination	–0.40	0.82	0.32	0.86
Peripheral vision	–0.23	0.70	0.18	1.01
Static acuity	–0.71	0.69	0.58	0.72

Note. Group centroids: low vision = .87, nonimpaired = 1.11.

experiencing significantly more difficulty with near acuity, distance acuity, and physical obstruction and that nonimpaired drivers had more trouble with motion perception.

DISCUSSION AND CONCLUSIONS

Results indicated that regardless of visual status, participants in this study reported visual difficulties in five specific areas while performing daily tasks. These factors indicated problems related to static acuity, dynamic acuity, peripheral vision, illumination, and contrast. Participants indicated that they experienced problems in reading small print; bumping into doorways, walls, and other things; recognizing and reading signs; and adjusting from bright to dim light and seeing indoors when the lights are dim. These findings, along with those of Kosnik et al. (1988, 1990) and Kline et al. (1992), suggest that both low-vision and nonimpaired older adults may experience problems related to contrast sensitivity, static and dynamic acuity, illumination, and peripheral vision simply because these problems are exacerbated by the aging process. If these findings are accurate, these problems will eventually affect the majority of older adults as they age, differing only in the degree of severity between low-vision and nonimpaired older adults. Although Kosnik et al. (1988) and Kline et al. (1992) did not specifically identify problems relating to static acuity and contrast sensitivity as being difficult for older adults, the variables that loaded on these factors in the present study were the same as variables that loaded on the factors that Kosnik et al. (1988) defined as "near acuity" and "clumsiness" (p. 64) and the factor that Kline et al. (1992) identified as "general vision problems associated with aging" (p. 31).

The problems of contrast sensitivity, static and dynamic acuity, and illumination may be interrelated. Owsley, Sekuler, and Siemsen (1983) suggested that acuity levels have a significant effect on contrast sensitivity and that individuals with poorer acuities tend to be less sensitive to low-contrast stimuli with high spatial frequencies.

TABLE 6: Summary of Discriminant Analysis for Experiences in Driving Predictors[a]

Predictor Variable	Standardized Function Coefficients	Unstandardized Function Coefficients	Pooled Within-Group Correlation Between Variables and Functions	Wilks's Lambda
Near acuity	0.41**	0.46	.76	.87
Peripheral vision	–0.22	–0.27	.12	1.00
Glare	–0.20	0.21	.27	.98
Physical obstruction	0.51*	0.60	.43	.96
Motion perception	–0.54	–0.68	–.09	1.00
Distance acuity	0.55**	0.69	.71	.89
Constant		0.06		

[a] χ^2 (6, N = 103) = 21.71, $p < .001$, Wilks's lambda = .50.

*$p < .05$. **$p < .001$.

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TABLE 7: Pooled Within-Group Correlation Matrix for Experiences in Driving Task LDF

Predictor	1	2	3	4	5	6
Distance acuity	1.00					
Glare	.47	1.00				
Motion perception	.33	.40	1.00			
Near acuity	.53	.26	.15	1.00		
Physical obstruction	.26	–.01	.43	.22	1.00	
Peripheral vision	.47	.36	.44	.12	.38	1.00

Illumination levels also affect contrast sensitivity. Owsley et al. (1983) asserted that reduced retinal illuminance, caused by a decreased pupil diameter or an opaque lens, also affects contrast sensitivity by selectively depressing sensitivity for higher frequencies. In light of the findings of this study, it seems reasonable to assume that the reported difficulties associated with acuity and contrast are related to the aging process and that reduced illumination (environmental and physiological in nature) serves only to exacerbate the problems.

Results from the factor analysis performed on the experiences while driving questions revealed that both low-vision and nonimpaired drivers reported visual difficulties in six areas: near acuity (with emphasis on night driving), peripheral vision, glare (with emphasis on night driving), visual obstruction, motion perception, and distance acuity. Examples of the types of driving-related tasks that the participants listed as troublesome include staying in their lane, seeing over the steering wheel or dashboard, and judging their speed without looking at the speedometer. Participants also noted that they often had trouble seeing the taillights of other vehicles, seeing through dirty windshields, keeping the instrument panel in focus, merging into traffic, and seeing at night because of glare from oncoming headlights.

Findings from this study were similar to those reported by Kline et al. (1992), who reported five specific age-related visual problems that affect the driving abilities of older adults. Interestingly, in the Kline et al. (1992) study, it can be noted that no factors relating to night driving or near acuity were identified; however, variables relating to night driving and near acuity both loaded on a factor that they referred to as an "age factor" (p. 31).

Do low-vision and nonimpaired adults differ in the magnitude or type of vision problems they experience when performing daily tasks? Results indicated that all five factors identified from the daily vision questions could be used in a linear discriminant function to differentiate low-vision and nonimpaired older adults. Low-vision adults reported experiencing more difficulties with tasks relating to static acuity. Nonimpaired older adults reported experiencing more difficulty with dynamic acuity and illumination. Therefore, both low-vision and nonimpaired older adults find tasks relating to static acuity, illumination, and

TABLE 8: Group Means and Standard Deviations for Experiences in Driving LDF

	Nonimpaired		Low Vision	
Predictors	*M*	*SD* *n*(103)	*M*	*SD* *n*(103)
Motion perception	0.00	0.70	–0.07	0.94
Distance acuity	–0.30	0.76	0.30	0.87
Glare	–0.15	0.83	0.11	1.12
Near acuity	–0.30	0.65	0.40	1.23
Peripheral vision	–0.10	0.82	0.00	0.79
Physical obstruction	–0.12	0.76	0.25	1.00

Note. Group centroids: low vision = 0.35; nonimpaired = 0.67.

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dynamic acuity as troublesome. However, low-vision older adults report experiencing unique difficulties with static acuity related tasks. Conversely, nonimpaired older adults find tasks related to illumination and dynamic acuity as uniquely troublesome. The effectiveness of the linear discriminant function was demonstrated by the fact that 80.6% of the low-vision participants and 87.1% of the nonimpaired participants were correctly classified into the appropriate groups.

The discriminant analysis performed on the questions about experiences while driving indicated that the factors of near acuity, distant acuity, and physical obstruction could be used in a LDF to differentiate low-vision from nonimpaired older adult drivers. Low-vision drivers reported greater difficulty performing driving tasks that involved near acuity, distance acuity, and physical obstruction. These difficulties included keeping the instrument panel in focus at night, seeing the taillights of oncoming vehicles, experiencing difficulty because of glare on the windshield, and difficulty seeing past the steering wheel or dashboard. Nonimpaired adult drivers reported experiencing more difficulty with tasks related to motion perception, such as judging their own speed without looking at the speedometer. This particular finding is interesting because the factor of motion perception did not contribute significantly to the linear discriminant function. However, nonimpaired adults distinctly reported motion perception tasks as troublesome for them while driving. These individuals may have developed coping mechanisms that allow them to effectively perform these tasks, although they still deem them troublesome. Unfortunately, the LDF produced from these factors was not as effective in distinguishing between the groups as the LDF produced by the daily task questions. The function correctly classified 73.1% of the nonimpaired drivers, but only 58.3% of the low-vision drivers were correctly identified.

Finally, it would be logical to assume that low-vision and nonimpaired older adults use different compensatory skills when driving to deal with the uniquely different visual problems that they experience. However, the two groups differed only in the amount of time that they spent driving at night. Low-vision older adults spent, on average, 0% to 5% of their total driving time at night; conversely, nonimpaired adults spent between 6% and 10% of their total driving time at night. The two groups did not differ significantly in the amount of time driving in rush hour traffic, the total number of miles driven, or the type of environments in which they chose to drive (e.g., driving in rural locations vs. highway driving).

PRACTICAL IMPLICATIONS

These data may prove beneficial to a variety of groups interested in issues related to aging and driving. For example, results indicated that nonimpaired drivers experienced difficulty judging their speed without looking at the speedometer. This could be addressed by providing drivers with the option of having an audible speedometer that provides updates about one's speed by simply pressing or programming a button on the steering wheel.

Designers may also consider providing low-vision drivers with the option of purchasing an instrument panel that magnifies the instrument readings, thus enlarging the displayed readings for better viewing. The combined alterations might assist low-vision drivers to better view their display panels when driving at night or during inclement conditions.

Automobile designers have already attempted to address the issue of physical obstructions by making automobiles that have tilting steering wheels. However, some participants in the study complained that the posts that hold the windshield in place also cause considerable obstruction of their view. Engineers may consider ways to reduce the size of the posts or to move them farther to the sides of the vehicles, out of the peripheral field of drivers, without compromising the integrity of the automobile. Finally, larger side mirrors may assist low-vision adult drivers with staying in their lane.

To address the issue of distance acuity, which was identified as problematic by many participants in this study, auto designers may consider coating the surface of windshields with a substance that better repels dirt, haze, and raindrops, thus enhancing the driver's visual abilities. A commercial product that performs this function is currently available; however, it requires

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continuous applications, a task that many older adult drivers may not remember to do.

Additionally, a sensor that detects the flash of brake lights from preceding cars and then sounds an alarm when one gets too close, similar to the rear-mounted sensors currently in some vehicles, may assist some older drivers who have difficulty seeing the brake lights of other cars.

Highway advisory boards may also find these data useful when planning and designing transportation infrastructures for an aging population. Distance acuity involves the ability to read street signs. Participants complained that they often fail to turn onto streets because they miss the street signs. Their inability to read the street signs may also be attributable to the size of the signs, the type and size of letters used on the signs, or the contrast between the letters and the sign itself. Additional research in these areas may assist in the design of more effective signs for use by older adult drivers.

Although the questions seem simple, the answers that may prove valuable in assisting automobile designers, highway advisory boards, and engineers in preparing for an aging population are not simple. The answers can be found only through continued research into these areas.

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Loretta Neal McGregor is the chairperson of the Department of Psychology and Counseling and an associate professor at Arkansas State University in Jonesboro, Arkansas. She obtained her Ph.D. in human factors psychology from Wichita State University in 2000.

Alex Chaparro is an associate professor of psychology at Wichita State University. He obtained his Ph.D. in experimental psychology from Texas Tech University, Lubbock, Texas, in 1990.

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